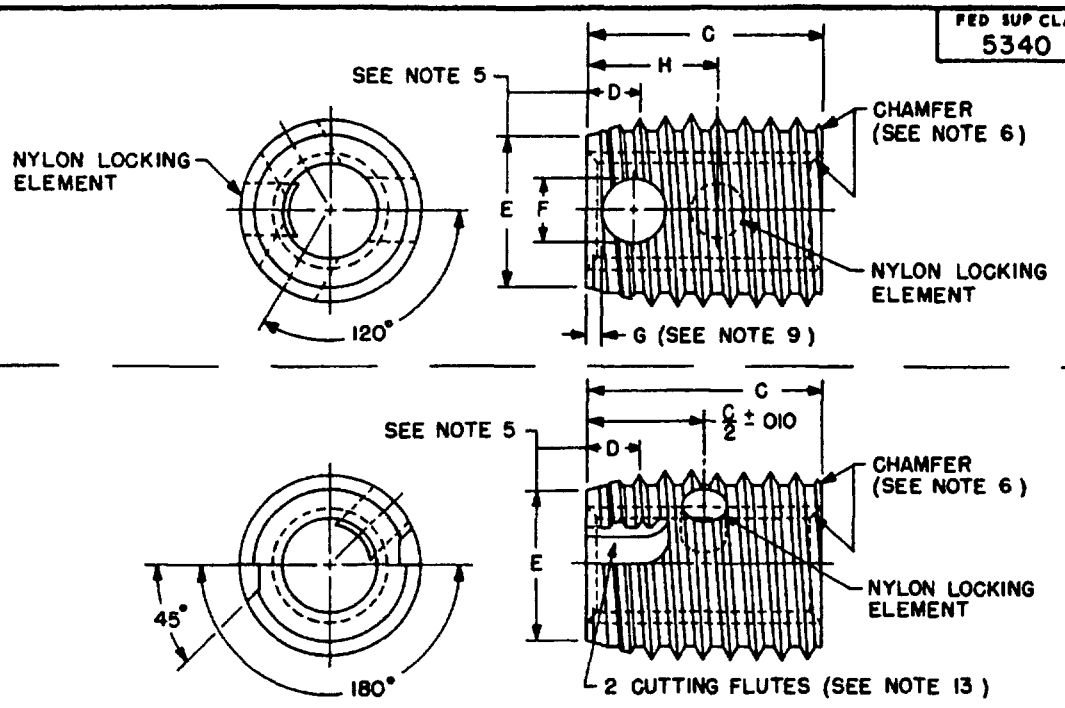
FED SUP CLASS
5340



OPTIONAL DESIGNS

NOTES:

(F) 1. MATERIAL: Carbon steel, AISI 1117, in accordance with Fed. Std. No 66. Corrosion-resisting steel, type 303 and 303MA, annealed condition, hot or cold finish (as rolled) in accordance with Fed. Std. No. 66.

2. HEAT TREATMENT: Carbon steel inserts shall be case hardened to a depth of .003-.005 inch with a hardness of Rockwell 15N75 minimum.

3. PROTECTIVE COATING: Carbon steel inserts shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3. Corrosion-resisting steel inserts shall be passivated in accordance with QQ-P-35.

4. THREADS: Internal threads shall be in accordance with Screw Thread Standards for Federal Services. Handbook H28. External threads shall be self-tapping and have a 60° thread approximating American National form.

5. DIMENSIONS: All dimensions are in inches. "D" dimension refers to the tapered portion of the external thread (pilot length). "E" dimension refers to the small end of the tapered portion of the external thread (pilot diameter).

6. CHAMFER: Chamfer internal and external threads to one thread length.

7. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS35914-125. For internal thread lock, add "L" after the dash number. Example: MS35914-125L.

8. LOCKING ELEMENT: Nylon locking element is available only for inserts having a specified "H" dimension.

9. "G" dimension equals approximately 60% of external thread pitch.

10. Thin wall inserts are supplied in corrosion-resisting steel only. No internal lock is available.

11 Referenced documents shall be of the issue in effect on date of invitations for bid.

12. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

13. Cutting flutes shall extend beyond the 2nd full thread.

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USER MI

APPROVED 16 OCT 1957 REVISED (A) 23 JUN 1960 (B) 23 APR 1965 (C) 25 APR 1969 (D) 21 JUL 1972 (E) 24 JUL 1975 (F) 20 SEPT 77

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P A. WC Other Cust 99 (F)	TITLE INSERT, SCREW THREAD-THREAD CUTTING	MILITARY STANDARD MS35914	
PROCUREMENT SPECIFICATION MIL-I-45916	SUPERSEDES	SHEET 1 OF 3	

DD FORM SEP 67 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5340-1217

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APPROVED 16 OCT 1957 REVISED (F) For changes see Sheet 1.

TABLE I
REGULAR WALL

FED SUP CLASS 5340

DASH NO.* CARBON STEEL	DASH NO.* COR RES STEEL	INTERNAL THREAD (-3B)	EXTERNAL THREAD (SEE NOTE 4)	C LENGTH ±.010	D MAX	D MIN	E DIA +.002 -.008	F DIA ±.010	H ±.010
125	141	.086-56UNC	9/64-48	.188	.050	.040	.123	.055	.109
201	251			.156	.050	.040		.055	.094
202	252			.125	.040	.032		.047	----
~~126~~	~~142~~	~~.086-64UNF~~	~~9/64-48~~	~~.188~~	~~.050~~	~~.040~~	~~.123~~	~~.055~~	~~.109~~
101	143	.112-40UNC	11/64-40	.234	.054	.044	.148	.062	.125
203	253			.187	.054	.044		.062	.115
204	254			.156	.049	.039		.055	----
~~103~~	~~144~~	~~.112-48UNF~~	~~11/64-40~~	~~.234~~	~~.054~~	~~.044~~	~~.148~~	~~.062~~	~~.125~~
103	145	.138-32UNC	7/32-32	.281	.068	.058	.191	.078	.156
205	255			.218	.068	.058		.078	.125
206	256			.187	.056	.048		.070	----
~~104~~	~~146~~	~~.138-40UNF~~	~~7/32-32~~	~~.281~~	~~.068~~	~~.058~~	~~.191~~	~~.078~~	~~.156~~
105	147	.164-32UNC	1/4-32	.328	.068	.058	.222	.078	.188
207	257			.250	.068	.058		.078	.135
208	258			.218	.056	.048		.070	----
~~106~~	~~148~~	~~.164-36UNF~~	~~1/4-32~~	~~.328~~	~~.068~~	~~.058~~	~~.222~~	~~.078~~	~~.188~~
107	149	.190-24UNC	19/64-24	.375	.092	.082	.259	.109	.219
108	150	.190-32UNF	19/64-24	.375	.092	.082	.259	.109	.219
211	261			.296	.092	.082		.109	.172
212	262			.250	.069	.059		.094	----
109	151	.250-20UNC	3/8-20	.484	.109	.097	332	.125	.281
110	152	.250-28UNF	3/8-20	.484	.109	.097	.332	.125	.281
215	265			.375	.109	.097		.125	.203
216	266			.312	.099	.089		.109	----
111	153	.3125-18UNC	15/32-18	.562	.123	.110	.420	.141	.312
112	154	.3125-24UNF	15/32-18	.562	.123	.110	.420	.141	.312
219	269			.469	.123	.110		.141	.250
220	270			.375	.113	.102		.125	----
113	155	.375-16UNC	9/16-16	.687	.137	.121	.510	.156	.375
114	156	.375-24UNF	9/16-16	.687	.137	.121	.510	.156	.375
223	273			.562	.137	.121		.156	.281
224	274			.437	.126	.113		.141	----
~~115~~	~~157~~	~~.4375-14UNC~~	~~41/64-14~~	~~.781~~	~~.160~~	~~.140~~	~~.581~~	~~.188~~	----
~~116~~	~~158~~	~~.4375-20UNF~~							
117	159	.500-13UNC	47/64-13	.906	.185	.160	.668	.219	.500
118	160	.500-20UNF	47/64-13	.906	.185	.160	.668	.219	.500
227	277			.750	.185	.160		.219	.391
228	278			.562	.165	.143		.188	----
~~119~~	~~161~~	~~.5625-18UNC~~	~~13/16-12~~	~~1.000~~	~~.198~~	~~.170~~	~~.742~~	~~.234~~	----
~~120~~	~~162~~	~~.5625-18UNF~~							
121	163	.625-11UNC	29/32-11	1.125	.213	.183	.827	.250	----
122	164	.625-18UNF	29/32-11	1.125	.213	.183	.827	.250	----
231	281			.937	.213	.183		.250	----
232	282			.687	.192	.166		.219	----
123	165	.750-10UNC	1-5/64-10	1.375	.240	.205	.993	.281	----
124	166	.750-16UNF	1-5/64-10	1.375	.240	.205	.993	.281	----
235	285			1.125	.240	.205		.281	----
236	286			.812	.208	.178		.234	----

* Lined out dash numbers are INACTIVE FOR NEW DESIGN AFTER 24 JUL 1975

INTERCHANGEABILITY

The slotted type inserts covered by dash numbers 1 thru 26 and 41 thru 66 given in previous revisions of this standard are cancelled after 23 April 1965 and superseded by dash numbers 101 thru 126 and 141 thru 166, respectively, given in the current revision of this standard.

P.A. WC Other Cust 99	TITLE INSERT, SCREW THREAD - THREAD CUTTING	MILITARY STANDARD MS35914
PROCUREMENT SPECIFICATION MIL-I-45916	SUPERSEDES	SHEET 2 OF 3

DD FORM 1 SEP 57 672-1 (COORDINATED)

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FED SUP CLASS 5340

TABLE II
THIN WALL

DASH NO. COR RES STEEL	INTERNAL THREAD (-3B)	EXTERNAL THREAD (SEE NOTE 4)	C LENGTH ±.010	D MAX	D MIN	E DIA +.002 -.005	F DIA ±.010
301	.112-40UNC	5/32-40	.234	.054	.044	.136	.062
302			.187	.054	.044		.062
303			.156	.046	.038		.055
304	138-32UNC	3/16-32	.281	.068	.058	.160	.078
305			.218	.068	.058		.078
306			.187	.064	.054		.070
307	.164-32UNC	7/32-32	.328	.068	.058	.191	.078
308			.250	.068	.058		.078
309			.218	.064	.054		.070
310	.190-32UNF	17/64-32	.375	.092	.082	.237	.109
311			.296	.092	.082		.109
312			.250	.084	.074		.094
313	.250-28UNF	11/32-28	.484	.109	.097	.310	.125
314			.375	.109	.097		.125
315			.312	.099	.089		.109
316	.3125-24UNF	13/32-24	.562	.123	.110	.368	.141
317			.469	.123	.110		.141
318			.375	.113	.102		.125
319	.375-24UNF	1/2-20	.687	.137	.121	.457	.156
320			.562	.137	.121		.156
321			.437	.126	.113		.141
322	.500-20UNF	5/8-18	.906	.185	.160	.576	.219
323			750	.185	.160		.219
324			.562	.165	.143		.188

TABLE III
RECOMMENDED HOLE SIZES

INTERNAL THREAD	REGULAR WALL: FOR HIGH STRENGTH LIGHT ALLOYS OF LESS THAN AVERAGE MACHINABILITY, CAST IRON, MALLEABLE IRON AND MILD STEEL		REGULAR WALL: FOR MEDIUM STRENGTH LIGHT ALLOYS OF AVERAGE MACHINABILITY, CAST IRON AND HIGH STRENGTH THERMOSETTING PLASTICS		REGULAR WALL: FOR LOW STRENGTH LIGHT ALLOYS OF EXCELLENT MACHINABILITY, THERMOPLASTICS AND THERMOSETTING PLASTICS		THIN WALL: FOR ALL STRENGTH ALLOYS, CAST IRON, MALLEABLE IRON AND ALL PLASTICS	
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN
.086	.134	.131	.131	.129	.129	.127	----	----
.112	.162	.159	.159	.156	.156	.152	.146	.140
.138	.207	.203	.203	.198	.198	.194	.175	.169
.164	.238	.234	.234	.230	.230	.226	.206	.200
.190	.281	.275	.275	.270	.270	.264	.252	.246
.250	.356	.349	.349	.343	.343	.336	.330	.322
3125	.447	.440	.440	.433	.433	.425	.390	.381
375	.538	.530	.530	.521	.521	.514	.479	.469
.4375	.613	.605	.605	.594	.594	.585	----	----
500	.704	.694	.694	.684	.684	.674	.605	.591
.5625	.780	.769	769	.759	.759	.748	----	----
.625	.871	.859	.859	847	.847	.835	----	----
.750	1.039	1.026	1.026	1.013	1.013	1.000	----	----

NOTE Hole preparation for locking inserts - Countersink to outside diameter of insert with 60° included angle.

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APPROVED 16 OCT 1957 REVISED (F) For changes see Sheet 1.

P A. WC	TITLE	MILITARY STANDARD
Other Cust 99	INSERT, SCREW THREAD - THREAD CUTTING	MS35914
PROCUREMENT SPECIFICATION MIL-I-45916	SUPERSEDES	SHEET 3 OF 3

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